Comparative Analysis of Clustering Algorithms for Epidemiological Population Modeling using the BRFSS Dataset

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***Abstract*—This paper presents a comparative unsupervised learning study using the Behavioral Risk Factor Surveillance System (BRFSS) dataset for epidemiological population analysis.**

Three clustering algorithms were evaluated:

- **K-Means**
- **DBSCAN**
- **Hierarchical Agglomerative Clustering**

The objective was to investigate how different clustering paradigms capture latent population structures, overlapping behavioral profiles, and heterogeneous public health patterns.

A preprocessing pipeline including categorical encoding, feature scaling, and Principal Component Analysis (PCA) was implemented before clustering.

Experimental results demonstrated that the epidemiological feature space exhibits predominantly continuous structures rather than sharply separated global groups. Among the evaluated methods, DBSCAN achieved the best structural representation by preserving dense continuous regions, identifying localized high-density subpopulations, and explicitly detecting sparse observations as noise.

The findings suggest that density-based clustering approaches are particularly suitable for modeling complex public health populations characterized by gradual transitions and overlapping demographic and behavioral attributes.

***Index Terms*—Machine Learning, Unsupervised Learning, Clustering, DBSCAN, K-Means, Hierarchical Clustering, Public Health, Epidemiology, BRFSS, PCA**

I. Introduction

Public health systems generate large-scale heterogeneous datasets containing demographic, behavioral, socioeconomic, and medical information. Extracting latent structures from such datasets is essential for identifying hidden population patterns, risk profiles, and epidemiological subgroups.

Traditional supervised learning methods require labeled data, which is often unavailable or difficult to obtain in epidemiological studies. Consequently, unsupervised learning methods such as clustering become valuable tools for exploratory analysis and population segmentation.

The Behavioral Risk Factor Surveillance System (BRFSS) dataset represents one of the largest public health survey systems in the United States. The dataset includes information related to:

- chronic diseases,
- behavioral habits,
- vaccination history,
- demographic attributes,
- socioeconomic conditions,
- and lifestyle indicators.

However, epidemiological datasets present several challenges:

- high dimensionality,
- mixed categorical and numerical variables,
- overlapping population profiles,
- nonlinear structures,
- and continuous transitions between health conditions.

This study evaluates three clustering paradigms under the same preprocessing pipeline:

1) K-Means (centroid-based clustering)
2) DBSCAN (density-based clustering)
3) Hierarchical Agglomerative Clustering

The central research question addressed in this work is:

"Which clustering paradigm best captures the latent epidemiological structure of the BRFSS dataset?"

The study compares the algorithms in terms of:

- structural behavior,
- geometric assumptions,
- density sensitivity,
- robustness to overlapping structures,
- and clustering quality metrics.

II. Methods and Materials

A. Dataset

The dataset used in this study corresponds to the Behavioral Risk Factor Surveillance System (BRFSS), a large-scale public health survey containing demographic, behavioral, and medical variables.

After preprocessing and feature selection, the final dataset included variables associated with:

- body mass index (BMI),
- smoking behavior,
- alcohol consumption,
- physical activity,
- chronic diseases,
- vaccination status,
- demographic characteristics,
- education level,
- and income groups.

The final processed matrix contained:

$$29999 \times 51 \tag{1}$$

features after encoding and transformation.

B. Preprocessing Pipeline

The preprocessing pipeline included:

1) Removal of irrelevant variables
2) Handling categorical variables
3) One-hot encoding
4) Feature scaling using StandardScaler
5) Principal Component Analysis (PCA)

PCA was applied to reduce dimensionality while preserving at least 90% of the variance.

The optimal number of PCA components was:

$$20 \tag{2}$$

C. K-Means Clustering

K-Means partitions observations into k centroid-based groups by minimizing the within-cluster sum of squares:

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2 \tag{3}$$

where:

- C_i represents cluster i
- μ_i is the centroid of cluster i

The optimal number of clusters was selected using silhouette analysis.

D. DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) identifies clusters as dense regions separated by sparse areas.

The algorithm uses two key hyperparameters:

- ε (eps): neighborhood radius
- min_samples: minimum density threshold

Unlike K-Means, DBSCAN:

- does not require specifying the number of clusters,
- handles irregular geometries,
- and detects outliers as noise.

E. Hierarchical Clustering

Hierarchical Agglomerative Clustering constructs nested cluster hierarchies by iteratively merging the closest groups.

Ward linkage was selected because it minimizes variance during cluster merging.

The linkage criterion is defined as:

$$D(A, B) = \frac{|A||B|}{|A| + |B|} ||\mu_A - \mu_B||^2 \tag{4}$$

where:

- A and B are clusters,
- μ_A and μ_B are centroids.

III. Results and Discussion

A. K-Means Results

Silhouette analysis selected:

$$k = 3 \tag{5}$$

as the optimal number of clusters.

The final metrics were:

TABLE I: K-Means Results

Metric	Value
Silhouette Score	0.1592
Davies-Bouldin Index	1.9164
Calinski-Harabasz Index	2874.74

The PCA projection revealed strong overlap between clusters, suggesting that the epidemiological space does not contain sharply separated spherical groups.

B. DBSCAN Results

DBSCAN achieved the best overall structural performance. The best configuration used:

$$\varepsilon = 2.9 \tag{6}$$

The resulting metrics were:

TABLE II: DBSCAN Results

Metric	Value
Silhouette Score	0.2047
Davies-Bouldin Index	1.5576
Calinski-Harabasz Index	267.32

DBSCAN identified:

- dense continuous regions,
- localized microclusters,
- and sparse anomalous observations.

Unlike K-Means, DBSCAN preserved the continuity of the epidemiological feature space rather than imposing rigid global partitions.

C. Hierarchical Clustering Results

Dendrogram analysis suggested that:

$$k = 4 \tag{7}$$

provided the best balance between interpretability and structural separation.

The final metrics were:

TABLE III: Hierarchical Clustering Results

Metric	Value
Silhouette Score	0.1691
Davies-Bouldin Index	1.7875
Calinski-Harabasz Index	465.52

Hierarchical clustering captured gradual transitions and local connectivity more effectively than K-Means, although the final partition still imposed discrete boundaries over a predominantly continuous structure.

D. Comparative Analysis

TABLE IV: Final Comparison of Clustering Algorithms

Algorithm	Silhouette	Davies-Bouldin	Main Strength
K-Means	0.1592	1.9164	Macrosegmentation
DBSCAN	**0.2047**	**1.5576**	Density awareness
Hierarchical	0.1691	1.7875	Hierarchical relations

The comparison revealed fundamentally different perspectives of the epidemiological space:

- K-Means imposed rigid centroid-based partitions.
- Hierarchical clustering preserved gradual relationships but still forced discrete cuts.
- DBSCAN provided the most flexible representation by preserving continuous dense regions and identifying sparse observations as noise.

E. Figures and Visualizations

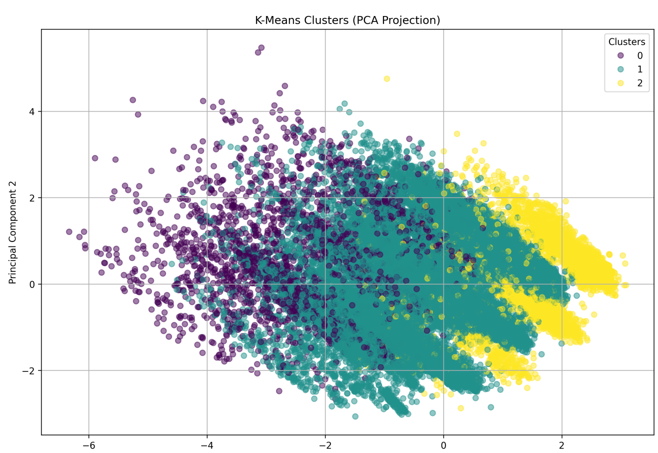

Fig. 1: PCA projection of K-Means clusters showing strong overlap between centroid-based partitions.

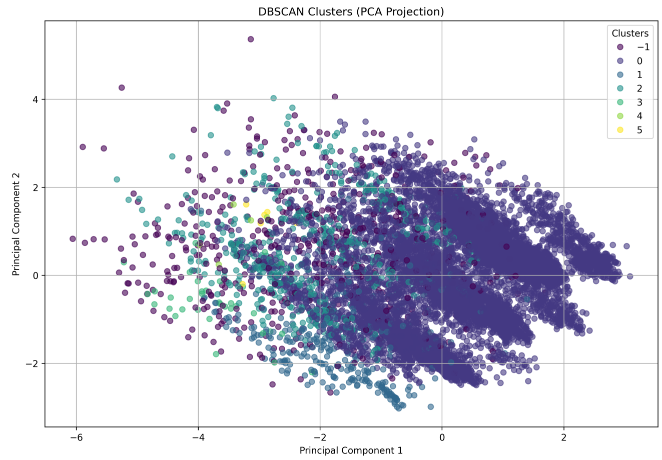

Fig. 2: DBSCAN PCA projection illustrating dense regions, local substructures, and noise observations.

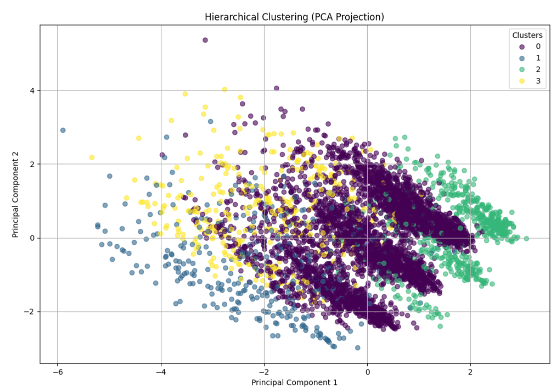

Fig. 3: Hierarchical clustering PCA visualization showing gradual transitions and chaining effects.

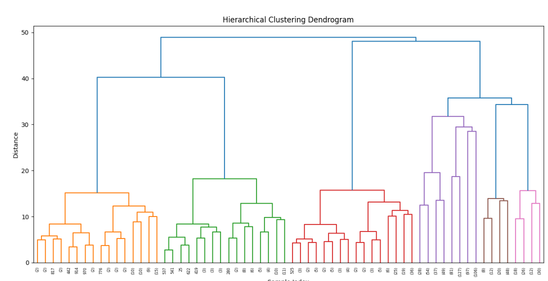

Fig. 4: Hierarchical clustering dendrogram used for selecting the number of clusters.

IV. Conclusions

This study evaluated three clustering paradigms for epidemiological population modeling using the BRFSS dataset.

The experimental results demonstrated that:

- the epidemiological feature space exhibits predominantly continuous structures,

- overlapping behavioral and demographic profiles complicate rigid segmentation,
- and density-based approaches provide more realistic representations of heterogeneous public health populations.

Among the evaluated methods, DBSCAN achieved the best structural representation by:

- preserving dense continuous regions,
- adapting to irregular geometries,
- detecting anomalous observations,
- and revealing localized epidemiological subpopulations.

The findings suggest that density-aware unsupervised learning approaches may be more suitable for modeling complex public health systems characterized by gradual transitions and overlapping health profiles.

Future work may explore:

- HDBSCAN,
- Gaussian Mixture Models,
- nonlinear dimensionality reduction,
- and temporal epidemiological analysis.

Acknowledgments

We thank the Machine Learning course instructors and the Universidad Distrital Francisco José de Caldas for academic guidance and computational resources.

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